Metals Exposure Registry and Investigation Tracking (MERIT) Specification

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Purpose and scope

The Metals Exposure Registry and Investigation Tracking (MERIT) system receives, stores and acts on laboratory results for lead and other metals for people of every age. Children, adults and pregnant people are first-class populations rather than variations of a child record.

It is built to be adopted by any U.S. state or local health jurisdiction. Anything that differs between jurisdictions, from reportable levels to the adult age cutoff, is a setting rather than code. Maryland examples appear only as examples.

In scope

- Lab results for any analyte in the analyte lookup table, starting with lead, mercury, cadmium and arsenic, in blood or urine.
- One person registry for children and adults. Population is derived from date of birth, collection date and pregnancy episodes.
- Residences, child care, schools and workplaces, each tracked over time.
- Case classification, case management, residential and occupational investigations, and prenatal follow-up.
- Federal extracts: quarterly files for CDC childhood blood lead surveillance (CBLS) and annual files for NIOSH adult surveillance (ABLES).

Out of scope: housing hazards unrelated to metals (mold, pests, radon), the lead rental registry, and billing or fee collection.

Why a new model. HHLPSS, built on California's RASSCLE II and the successor to STELLAR, assumes the subject is a child. That shows up in three places: guardian fields on the person, a single address, and an investigation that always targets a home. Adult and heavy-metal records get forced into that shape. One example is New York instructing labs to put a worker's employer in the guardian field.

Design principles

1. Population is derived from data, never set with a checkbox.

2. People link to people, and people link to organizations. The two kinds of link are never mixed.
3. Places are shared records. People attach to them by role and date range.
4. Every rule value is effective-dated data, not code. This covers thresholds, age cutoffs, units, lookback windows, environmental standards and federal extract layouts.
5. The as-reported lab record is never changed. Corrections live on the linked records, and a lab's own correction arrives as a new as-reported record that supersedes the old one.
6. Every create, update, merge, view and export of a person, result or case is audited.
7. Automation proposes and staff decide. Cross-links, clusters, merges, address changes and closures triggered by other events create review tasks; they never act on their own.

Glossary

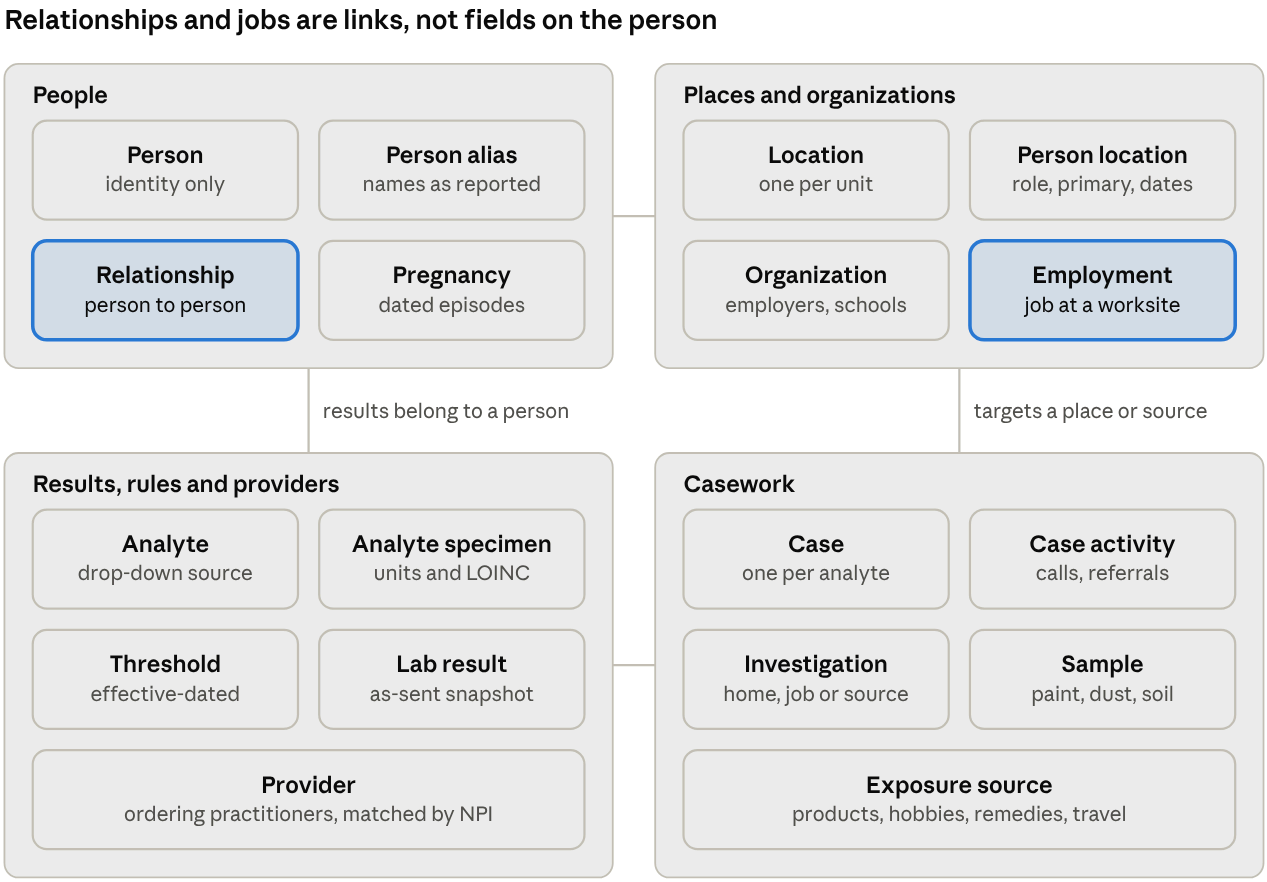
Term	Meaning in this spec
Analyte	The substance measured: lead, mercury, cadmium, arsenic, and any later addition.
Specimen	What was tested: whole blood or urine.
Sample type	How a blood specimen was drawn: venous, capillary or unknown. Applies to blood only.
Test method	How the lab measured the result, such as ICP-MS, graphite furnace AA or a point-of-care analyzer. Decides whether a venous result can confirm a case.
Speciation	The chemical form reported, such as total vs. inorganic arsenic or total vs. methyl mercury.
Population	Child, adult or pregnant; unknown when DOB is missing. Derived per result from date of birth, collection date and pregnancy episodes.
Primary residence	The single residence used for jurisdiction and federal reporting on a given date.

Term	Meaning in this spec
Reportable level	The value at or above which a lab must report. Some analytes are reportable at any value.
Action level	A value that triggers a workflow step, such as opening a case or scheduling a retest. Action levels are tiered, so several can apply to one analyte at once.
Case	One person's elevated exposure to one analyte, from opening to closure.
Investigation	A site visit or inquiry about one target: a home, a child-care site, a workplace or a non-occupational source. One investigation can serve several cases.
Take-home exposure	Metal carried from a workplace into a home on clothing, skin or vehicles.
Contact-only person	A person record with no lab results, such as a guardian or a household member.
ELR	Electronic laboratory reporting, normally HL7 v2.5.1 ORU^R01 messages.
eICR	Electronic initial case report sent from a provider's electronic health record through electronic case reporting (eCR).
LOINC	Standard codes identifying a lab test, used to map incoming results to an analyte and specimen.
NAICS	North American Industry Classification System; codes an employer's industry.
COC	Census occupation code; codes a worker's job.
CBLS	CDC childhood blood lead surveillance quarterly extract.
ABLES	NIOSH Adult Blood Lead Epidemiology and Surveillance annual data submission.

Term	Meaning in this spec
OSHA or state plan	Federal OSHA, or the state's own occupational safety agency in states that run an approved plan (for example, MOSH in Maryland).
Intake profile	A sender's saved description of its own file layout, used to parse that sender's files.
Held record	An incoming row that could not load on its own; it waits in a queue with named reasons.
Tenant	An agency or program that uses the system. Tenants form a tree under a root tenant, normally a state.
Cluster	A group of cases flagged for staff review because they share a place, employer, provider or product.

Data model overview

The core model has 18 entities in four groups. The person record carries identity only; guardians, addresses and employers are separate linked records with their own dates.



data model · 18 core entities in 4 groups

Person location ties people to places, and employment ties people to organizations at a worksite. Each lab result belongs to one person and one analyte specimen. A case groups one person's results for one analyte. Each investigation targets exactly one place, worksite or exposure source, and links to every case it serves.

Supporting tables are not drawn: contact methods, race values and external identifiers on the person; organization sites and location owners; the investigation-case link; environmental standards; intake batches, intake profiles and staging rows; consumer products, clusters and custom fields; and the tenant and security tables described under Architecture.

Entity definitions

Every table carries `id` (UUID), `created_at`, `created_by`, `updated_at`, `updated_by` and `row_version` for optimistic locking. Tenant-owned tables also carry `root_tenant_id` and `owning_tenant_id` (see Architecture). These are not repeated below. "Req" means

required to save; "Ext" means required before the record can go into a federal extract.

Enumerations come in two kinds. Values that drive program logic (population, statuses, operators, comparators, hold reasons) are system-defined. Descriptive values (roles, sources, closure reasons, sample types, source types) are lookup tables that administrators maintain, each mapped to the extract codes it feeds.

Person

The person record holds identity only. It has no address, guardian or employer fields.

Field	Type	Req	Rules
last_name	text(60)	Y	Trimmed; stored as reported and as a normalized match key.
first_name	text(60)	Ext	Labs sometimes send "BABY BOY"; flag these for review rather than reject.
middle_name	text(60)		
dob	date	Ext	Not in the future, not before 1900. Missing DOB makes population unknown and holds the person out of population rules.
dob_verified	boolean		True once staff confirm against a second source.
sex	enum	Ext	Female, male, other, unknown; mapped to each extract's codes.
race	multi-enum		Child table; any number of values from the race lookup. The lookup includes other, refused and unknown, so all eight ABLES race flags (Race1–Race8) can be derived.
ethnicity	enum		Hispanic, not Hispanic, unknown.
preferred_language	enum		Drives letter templates.
phone, email	text		Validated format; multiple values in a contact-method child table.

Field	Type	Req	Rules
identifiers	child table		External IDs as reported, such as a medical record number with its assigning facility or a Medicaid ID, each with type, source and date. Used by matching; shown only with person.view_identifiers.
deceased_date	date		On or after dob. Creates a task listing open cases to close with reason "deceased"; staff confirm.
merged_into_id	FK person		Set when the record is merged; merged records are read-only.

Person alias

Names as they arrived on lab reports (nicknames, maiden names, misspellings, placeholders such as "BABY BOY"), each with source and date. Used by matching; never shown as the primary name.

Field	Type	Req	Rules
person_id	FK person	Y	
last_name, first_name, middle_name	text(60)	Y (last name)	As reported, plus normalized match keys.
alias_type	enum	Y	As reported, nickname, former name, placeholder, misspelling.
source	enum	Y	Lab report, eICR, interview, staff.
source_result_id	FK result		The result that carried this name, when there is one.
first_seen, last_seen	date	Y	last_seen updates each time the same name arrives again.

Pregnancy episode

Field	Type	Req	Rules
person_id	FK	Y	Warn, rather than block, when the person's sex is male; staff confirm.
start_date	date	Y	Estimated conception or first known pregnant date.
expected_end_date	date		
actual_end_date	date		On or after start_date.
outcome	enum		Live birth, loss, unknown. A live birth prompts linking the newborn.
source	enum	Y	Lab report, eICR, provider, interview.

A result collected during an open episode takes the pregnant population. An episode is open from start_date through actual_end_date; with no actual end, through expected_end_date; with neither, for the configured maximum length (default 42 weeks). A work queue lists open episodes past their expected end.

Person relationship

Links a person to another person. This replaces guardian fields. The role says what the related person is to the person: role parent means the related person is this person's parent, and the related person's page shows the inverse label (child of).

Field	Type	Req	Rules
person_id	FK person	Y	
related_person_id	FK person	Y	Not the same as person_id. Related person may be contact-only.
role	enum	Y	Parent, legal guardian, foster parent, other caregiver, spouse or partner, sibling, household member, emergency contact. Each role has an inverse label.

Field	Type	Req	Rules
is_primary_contact	boolean		At most one current primary contact per person.
has_legal_custody	boolean		
start_date, end_date	date		end_date ⩾ start_date; null end = current.
source	enum	Y	Lab report, eICR, interview, case manager, other agency.
verified_date	date		Set when staff confirm a relationship that arrived on a report.

Organization

Employers, child-care providers, schools, labs, provider practices and health departments. Individual practitioners are Provider records, not organizations.

Field	Type	Req	Rules
name	text(120)	Y	Duplicate warning on normalized name plus any shared site address.
org_type	multi-enum	Y	One organization can be several types, such as a school that is also an employer.
naics_code	text(6)	Ext for employers	Validated against the NAICS lookup for the stored code version (NAICS 2022 for ABLES today); 999 = unknown.
npi	text(10)		NPI-2 for practices and labs; check digit validated (see Provider and laboratory reference data).
clia_number	text(10)	For labs	Format checked: two digits, the letter D, seven digits.
parent_org_id	FK org		Corporate parent of a site-level employer.

Field	Type	Req	Rules
contact_name, phone, email	text		
active	boolean	Y	

Sites are LOCATION records linked through an organization-site table (org_id, location_id, site_name, start/end).

Location

One record per physical address and unit, shared by everyone linked to it. Addresses outside the US and city-only locations, for people without a fixed address, are allowed.

Field	Type	Req	Rules
country	text(2)	Y	ISO 3166 code; defaults to US. ABLES codes Canada and Mexico separately.
address_line1	text	Y	Optional only for a city-only location.
unit	text		Part of identity: two apartments are two locations. The row with no unit is the building record.
city	text	Y	
state, zip5	text	Y for US	State is a 2-letter code; zip5 is 5 digits.
region, postal_code	text		For addresses outside the US.
county_fips	text(3)	Ext	From geocoding; 999 = unknown.
std_status	enum	Y	Standardized, standardized with changes, failed, not attempted.
latitude, longitude	decimal		Set by the geocoder, or by staff placing a pin in the field.
geocode_quality	enum		Rooftop, parcel, street, ZIP centroid, manual pin.

Field	Type	Req	Rules
census_tract	text(11)		Stored with the census vintage it came from (2020 today).
building_id	FK location		The building record: same address, no unit. Groups units for cluster detection and common-area investigations.
year_built	int		1700 to current year; source recorded. Built before 1978 shows as a risk flag.
housing_type	enum		Single-family, multi-unit, mobile home, institutional, commercial.
tenure	enum		Owner-occupied, rental, unknown.
owners	child table		Owner person or organization (exactly one per row), with start and end dates and source, so notices and orders go to the owner at the time.

Person location

Where a person lives or spends time. Rules in the next section.

Field	Type	Req	Rules
person_id, location_id	FK	Y	
role	enum	Y	Residence, child care, school, relative or sitter home, other regular site.
is_primary	boolean	Y	Residence role only.
primary_reason	enum	When overlapping	Custody order, most overnights, guardian designation, lab-reported address, staff determination.
schedule_type	enum		Full time, weekdays, weekends, alternating weeks, part time, seasonal.

Field	Type	Req	Rules
hours_per_week	decimal		0 to 168. Compared with the configured minimum when deciding which places get an investigation.
schedule_note	text		
start_date, end_date	date		Null end = current.
source	enum	Y	Lab report, eICR, interview, case manager, other agency.
verified_date	date		

Employment

A person's job at an organization. A person can hold several at once.

Field	Type	Req	Rules
person_id	FK	Y	
org_id	FK org	Y	Self-employed workers get an organization of type self-employed.
worksite_location_id	FK location		Must be one of the organization's sites, or a job site such as a construction address.
job_title	text		As stated.
industry_text	text		Industry as stated; kept for coding.
occupation_code	text(4)	Ext if occupational	Census occupation code, stored with its code version. ABLES currently expects Census 2002 codes, with 990 = unknown.
naics_override	text(6)		When this job's industry differs from the organization's.

Field	Type	Req	Rules
metal_tasks	text		Description of tasks involving metals.
analytes_exposed	multi FK analyte		
ppe_notes	text		Respirator, change rooms, laundering.
start_date, end_date	date		Null end = current.
source	enum	Y	Lab report, eICR, interview, employer, case manager.
verified_date	date		Set when staff confirm a job that arrived on a report.

Exposure source

Non-occupational sources, for any population.

Field	Type	Req	Rules
person_id	FK	Y	
case_id	FK		
source_type	enum	Y	Renovation, hobby, casting, firearms or range, consumer product, spice or food, traditional remedy, cosmetic, cookware, pica, retained bullet fragment, travel, water, other. Each type maps to an ABLES Process code.
description	text		Product name, brand, country of origin. Sent as the ABLES Process narrative, which NIOSH prefers to codes.
product_id	FK product		Links to the product registry when the source is a product.
location_id	FK		Where the exposure happens, if somewhere other than the residence.

Field	Type	Req	Rules
start_date, end_date	date		
confirmed	boolean		True when a sample confirms it.

Analyte, analyte specimen and threshold

Defined in the section on analytes and thresholds.

Lab result

Field	Type	Req	Rules
person_id	FK	Y	Set by matching; null while in the unmatched queue.
analyte_specimen_id	FK	Y	From LOINC mapping or the analyte and specimen dropdowns.
collected_date	date	Y	⩾ dob; ⩽ reported_date when present, and ⩽ received_at.
analyzed_date	date		Between collected and reported.
reported_date	date		When the lab released the result.
received_at	timestamp	Y	When the agency received it; taken from the intake batch.
result_status	enum	Y	Preliminary, final, corrected or cancelled, as the lab sent it.
supersedes_result_id	FK result		Set when a corrected or cancelled result replaces an earlier one with the same accession.
value_reported	decimal	Y*	*Or value_text when the lab sends a non-numeric result.

Field	Type	Req	Rules
operator	enum		=, <, >. A "<" result is stored as below detection with the limit as value.
unit_reported	text	Y	Must be in the analyte specimen's accepted units.
value_canonical, unit_canonical	decimal, text	Y	Computed at intake; read-only.
sample_type	enum	Blood only	Venous, capillary, unknown. Must be null for urine. Taken from the LOINC code when the code implies it.
test_method	enum		ICP-MS, graphite furnace AA, anodic stripping voltammetry, point-of-care analyzer, other, unknown. Used by classification.
device_model, device_serial, kit_lot	text		When known; recall queries use them.
speciation	enum	When allowed	From the analyte specimen's speciation options.
creatinine_value	decimal	When required	Needed to compute a creatinine-adjusted urine value.
value_creatinine_adj	decimal		Computed.
reason_for_test, payer_type	enum		As sent. Reason: screening, confirmation, follow-up, occupational monitoring. Payer: Medicaid, private, uninsured, unknown.
lab_org_id	FK org	Y	
ordering_provider_id	FK provider		Linked by NPI or matching; the reported name stays in the snapshot.

Field	Type	Req	Rules
ordering_facility_id	FK org		
accession_number	text		Part of duplicate detection.
loinc_reported	text		As sent.
reported_snapshot	JSON	Y	Every patient, guardian, employer and address field exactly as received. Never edited.
reported_location_id	FK location		The geocoded as-reported address.
age_at_collection	interval		Computed.
population	enum		Computed: child, adult, pregnant, unknown.
status	enum	Y	Valid, duplicate, superseded, entered in error, rejected.
intake_batch_id	FK	Y	Points to the raw message or file.

Duplicates and corrections. Same person, analyte specimen, collected date, lab and accession number (or same value when accession is missing) is a duplicate. A record that arrives with a corrected or cancelled status is not a duplicate: it supersedes the earlier result, which is marked superseded, and classification reruns. Preliminary results can trigger action-level tasks but never count toward classification or federal extracts; the final result supersedes them.

Case

Field	Type	Req	Rules
person_id	FK	Y	
analyte_id	FK	Y	One open case per person per analyte.
population	enum	Y	Set from the qualifying result; changes only by staff action, with a reason.

Field	Type	Req	Rules
classification	enum	Y	Confirmed, suspect, not a case (after review).
status	enum	Y	Open, monitoring, closed.
qualifying_result_id	FK result	Y	
highest_result_id	FK result		Updated as results arrive.
exposure_category	enum		Residential, occupational, both, other non-occupational, unknown. Maps to ABLES WorkRel.
assigned_to	FK user		
opened_date, closed_date	date		
closure_reason	enum	On close	Below level on retest, lost to follow-up, moved out of state, declined, deceased, duplicate, other.

Case activities (a child table) hold contact attempts, education, referrals, medical management notes and letters, each with type, date, worker, outcome and note. A case-cluster table links cases to clusters.

Investigation

Field	Type	Req	Rules
inv_type	enum	Y	Residential, child-care or school, workplace, non-occupational source.
location_id	FK location	One of	Residential and child-care or school types; also a source investigation at a place, such as a firing range.
org_id, worksite_location_id	FK	One of	Workplace type: the employer and the worksite.

Field	Type	Req	Rules
exposure_source_id	FK	One of	A source investigation with no fixed place, such as a consumer product.
status	enum	Y	Requested, scheduled, in progress, findings issued, remediation, cleared, closed.
assigned_to	FK user		
requested, scheduled, completed dates	date		In order.
findings	text		
regulatory_action	enum		None, notice, order, referral to OSHA or state plan, referral to another agency.
referral_date, outcome	date, text		

Exactly one target is set, matching the type. An investigation-case link table ties each investigation to every case it serves, and to the employments that brought a workplace into scope; an investigation started from a location or cluster can have no case. Before creating one, the system offers any open investigation of the same target, so siblings in one home share one investigation.

Environmental samples (a child table) hold sample type (XRF paint, dust wipe, soil, water, product, air, surface wipe), room or component, value, unit and the standard compared against. Standards come from an effective-dated environmental standards table (sample type, surface, value, unit, authority, dates), and exceeds_standard is computed, not typed. EPA's 2024 dust rule shows why: it replaced fixed dust-lead hazard standards with any reportable level and set post-abatement action levels of 5, 40 and 100 µg/ft² for floors, window sills and troughs (EPA). Instrument readings, such as XRF exports, load through an intake profile instead of being retyped.

Address and primary residence rules

Each person has exactly one primary residence on any date they have a residence. Every other place they live or spend time is still recorded, and is investigated when it meets the rules below.

Rules

1. **One primary at a time.** No two person-location rows with role residence and is_primary true may overlap in date range for the same person. Enforce in the stored procedure that saves person locations, inside a transaction that locks the person's rows, not only in the UI.
2. **No gaps once known.** If a person has any residence covering a date, one of them must be primary for that date. Saving a second concurrent residence forces the user to pick the primary.
3. **Reason required on overlap.** When two residences overlap, primary_reason is required. Typical values: custody order, most overnights, guardian designation.
4. **Default when unknown.** With no other information, the most recent lab-reported address becomes primary with reason "lab-reported address" and verified_date empty. A work queue lists unverified primaries on open cases.
5. **New addresses from labs go to review.** When a new result arrives with an address different from the current primary, the system creates an address review task. It does not move the primary automatically. This behavior is a configuration setting.
6. **Two counties, two jobs.** The reporting county is the county of the primary residence on the collection date; it drives federal extracts and surveillance counts. The assignment county is the county of the current primary residence; it drives case assignment and local access. If no residence is known, both use the result's as-reported address.
7. **Snapshots never change.** Correcting a person's address history never edits the result's reported_snapshot or reported_location_id.
8. **Workplaces live on employment.** A work address is the worksite on an employment row, not a person-location row. This keeps residence history and job history separate and avoids double entry.
9. **Child care and school link to an organization.** The person-location row points to the organization's site, so several children at one provider share one location.
10. **Adults.** Multiple residences are allowed (seasonal homes, shelters) but the UI warns, rather than blocks, when an adult has two concurrent residences.
11. **Investigation scope.** Investigations draw on all locations active within a configurable lookback window before the qualifying collection date, not only the primary. Residences and child-care sites always get an investigation request. Other places get one when their hours per week meet the configured minimum, and a review task otherwise.

12. **Changing the primary splits rows.** The old primary row ends the day before the new start date. If the person still lives there, a new non-primary row continues from that date with the same schedule, so history stays exact.
13. **No fixed address.** A person without a fixed address gets a residence on a shelter's location or a city-only location, so jurisdiction still resolves.

Example: shared custody

Location	Role	Schedule	Primary	Reason
Mother's home	Residence	Weekdays	Yes	Custody order
Father's home	Residence	Weekends	No	
ABC Child Care	Child care	Weekdays, 40 h/week	n/a	
Grandmother's home	Relative or sitter home	Two afternoons a week	n/a	

The county of the mother's home is used for reporting and assignment. Residential investigation requests are generated for both homes and the child-care site, in that order. The grandmother's home gets a request if two afternoons a week meets the configured minimum hours; otherwise it gets a review task.

Analytes, thresholds and result normalization

Analytes are rows in three lookup tables, so adding a metal or changing a level is an administrator task, not a software release.

Analyte

Field	Type	Rules
code	text(8)	Unique: PB, HG, CD, AS.
display_name	text	Shown in drop-downs: Lead, Mercury, Cadmium, Arsenic.

Field	Type	Rules
active	boolean	Inactive analytes stay on old records but leave the drop-down.
sort_order	int	Drop-down order.
extract_targets	multi-enum	CBLS, ABLES, state report, none.

Analyte specimen

One row per analyte and specimen pair. This table drives which fields the result form shows.

Field	Type	Rules
analyte_id	FK	
specimen	enum	Whole blood, urine.
canonical_unit	text	Unit stored for comparison, such as µg/dL for blood lead.
accepted_units	child table	Unit text as labs send it, plus multiplier to canonical. A unit not listed sends the result to the error queue.
loinc_codes	child table	Codes that map to this pair, each with any sample type or speciation the code implies. For blood lead, LOINC 5671-3 is discouraged in favor of 77307-7 (venous) and 10368-9 (capillary). Unknown codes go to the error queue.
uses_sample_type	boolean	True for blood lead. Shows the venous/capillary field.
requires_creatinine	boolean	Shows the creatinine field and computes an adjusted value.
speciation_options	multi-enum	For example, arsenic in urine: total, inorganic, organic. Empty hides the field.

Threshold

Field	Type	Rules
analyte_specimen_id	FK	
population	enum	Child, adult, pregnant, all.
age_min, age_max	interval	Optional bounds within a population, for laws that set levels by age band.
speciation	enum	Optional; a level can apply only to inorganic arsenic, for instance.
sample_type	enum	Optional; a level can apply only to venous or only to capillary results.
threshold_type	enum	Reportable, case, action.
action_code	FK	The workflow step this level triggers, such as "retest due" or "open investigation".
due_interval	interval	For action rows: when the triggered step is due, such as 3 months.
value, unit	decimal, text	Compared in canonical units.
comparator	enum	⩾ or >.
effective_from, effective_to	date	No overlaps for the same analyte specimen, population, age band, speciation, sample type, type and action code, so tiered action levels run side by side.
authority	text	Regulation or guidance it comes from.
active	boolean	Reference rows ship inactive until an adopter turns them on.

The current CDC blood lead reference value for children, 3.5 µg/dL, is the first child lead row. All other levels, including each state's reportable levels for mercury, cadmium and arsenic, are loaded from the adopting jurisdiction's regulations rather than written into this spec. Federal reference rows ship inactive with their authority: the 2023 national case definition (3.5 µg/dL venous, all ages) and NIOSH's adult elevated level (5 µg/dL).

Action levels are tiered. CDC's schedule for a confirmatory venous test after an elevated capillary result is one example: within 3 months at 3.5–9 µg/dL, 1 month at 10–19, 2 weeks at 20–44 and 48 hours at 45 or more. Each tier is its own action row with its own due interval.

Intake normalization

Steps 1 to 5 run on every staging row before matching. Steps 6 and 7 need the matched person's DOB and pregnancy episodes, so they run after matching (see Intake pipeline).

1. Map LOINC (or the form's analyte and specimen) to an analyte specimen row, taking any sample type or speciation the code implies.
2. Parse the operator and numeric value. Keep the raw text.
3. Convert to the canonical unit using the accepted-unit multiplier.
4. For blood, store sample type; for urine, store null.
5. When creatinine is required and present, compute the adjusted value.
6. Derive age at collection and population from the matched person, falling back to the reported DOB.
7. Compare against effective thresholds for the collection date and run case rules.

Below-detection results. A "<" result is below a threshold only when its limit is at or below that threshold. When the limit is above it, as with a "<5" result against a 3.5 level, the result is indeterminate: it opens no case, joins a "detection limit above level" queue for follow-up with the reporter, and counts as incomplete in completeness reports. A ">" result is at or above a threshold when its limit is.

Arsenic note. Total urine arsenic is often raised by harmless organic arsenic from seafood. A total arsenic result above the action level should create a "speciation recommended" task rather than a confirmed case, unless the jurisdiction configures otherwise.

Case classification

Classification runs per result, per analyte, using the population derived for that result and the thresholds in force on its collection date.

Population derivation

1. No DOB → population unknown; the result goes to a review queue and opens no case automatically. If the value is at or above the lowest case level for any population, the review task is marked urgent.
2. Open pregnancy episode on the collection date → pregnant.
3. Age at collection below the configured adult cutoff → child.
4. Otherwise → adult.

The adult cutoff is a setting because the sources disagree. ABLES and the national case definition treat people 16 and older as adults, while some state reporting laws define children as 0–18 (Maryland's, for example). Federal extracts use their own age rules regardless of this setting, computed from DOB and collection date.

Child lead

Follows CDC's Lead in Blood 2023 case definition (CSTE 22-EH-01), with the level taken from the threshold table:

- **Confirmed:** one venous result at or above the level, measured by a confirming method, or two capillary results at or above it collected within 12 weeks of each other.
- **Suspect:** one capillary result at or above the level, or two capillary results at or above it collected more than 12 weeks apart.
- Unknown sample type counts as capillary for children, as the definition specifies. A venous result from a method that does not confirm also counts as capillary.
- Confirming methods are a setting. The default follows the definition, where graphite furnace AA and ICP-MS confirm; a result with no reported method confirms unless the jurisdiction configures otherwise.
- A suspect case opens with a "confirmatory venous test due" task, with the due date set by the action tier the result falls in.
- For annual counts and CBLS, a confirmed case counts as new once per calendar year unless it was confirmed in the previous year. When the two capillary results straddle a year end, it counts in the year of the first.

Adult lead

- A result at or above the adult case level opens a case. Unknown sample type counts as venous for adults, as the national definition specifies. A capillary result opens a suspect case with a confirmatory venous test task.

- ABLES Status follows NIOSH's own rules, not the case level. The extract takes the person's highest result in the calendar year and compares it with 10 µg/dL: at or above, the person is a new or existing case depending on the prior year's registry record; below, a new or existing adult (codes 1 to 5, or 9 when unknown).
- Exposure category stays unknown until the worker interview; the extract codes it as unknown until then.

Other metals

- A result at or above the case level for its analyte, specimen, speciation and population opens a case for that analyte.
- Mercury, cadmium and arsenic cases are separate from any lead case for the same person, but appear together on the person page.

General rules

- One open case per person per analyte. A new qualifying result attaches to the open case and may upgrade suspect to confirmed.
- Re-run classification when a result is corrected or superseded, a DOB changes, sample type or method changes, a merge happens, or a pregnancy episode is added, changed or removed. Show the user what changed.
- A threshold change never reclassifies closed cases, and open cases keep their classification; new results are evaluated under the new rows. A one-time report lists people whose recent results would qualify under the new level, for outreach. Reports for past periods can be rerun under either rule set.
- Staff can override a classification with a required reason; the computed value is kept beside it.

Investigation workflows

The case's population picks a workflow template; templates are configurable lists of steps, each with an owner role, a due-date rule and completion criteria. Three default templates ship: child residential, adult occupational and pregnancy. They differ in who is contacted and what gets investigated. Closure criteria, such as a number of results below the case level a minimum time apart with all investigations cleared, are template settings.

Child: residential workflow

1. **Open and assign.** Assign a case manager by assignment county (the current primary residence). Notify the local health department if configured.
2. **Confirm.** Track the confirmatory or follow-up test due date from the action tier.
3. **Reach the guardian.** Log each contact attempt against the person's primary contact relationship, or, if none is recorded yet, against the guardian created from the lab report at intake. Send a letter after the configured number of failed attempts.
4. **Exposure interview.** Record all residences and their schedules, child care, school, relative homes, household members' jobs, consumer products, remedies, pica, travel and any residence outside the US.
5. **Investigate each site.** Create one investigation per qualifying place (address rule 11), primary residence first, reusing any open investigation of the same place.
6. **Sample and act.** Record environmental samples, hazards found, notices or orders to the owner, remediation and clearance.
7. **Medical follow-up.** Record referrals (early intervention, nutrition, provider) and follow-up tests.
8. **Close.** Close when closure criteria are met, with a required reason.

Adult: occupational workflow

1. **Open and assign.** Assign an occupational investigator.
2. **Reach the worker.** Contact the person directly. No guardian step.
3. **Worker interview.** Record every current job with tasks, PPE and hours; non-occupational sources; pregnancy; and children in the household.
4. **Code the work.** Link or create each employer organization and worksite. Assign NAICS and occupation codes from the job title and industry text; an optional autocoding service can suggest codes for staff to accept.
5. **Employer follow-up.** Contact the employer or refer to the configured occupational safety agency, then track the referral outcome. Several workers at one worksite share one workplace investigation.
6. **Retest tracking.** Track repeat results against configured medical-removal and return-to-work levels.
7. **Close.** Close with a required reason.

An adult with a non-occupational source, such as renovation or a firing range, uses a non-occupational source investigation; if the source is in the home, a residential investigation is opened instead.

Pregnancy workflow

CDC's guidelines for pregnant and lactating women set their own follow-up levels, so this template has its own rows in the threshold table.

1. **Open and assign.** Assign a case manager by assignment county, and notify the prenatal care provider.
2. **Reach the person.** Contact the pregnant person directly.
3. **Exposure interview.** Record jobs and take-home exposure, residences, renovation, pica, consumer products, remedies, cookware and food from abroad.
4. **Investigate.** Open residential, workplace or source investigations as the interview points, under the same rules as the other templates.
5. **Retest tracking.** Track follow-up results during pregnancy against the pregnancy action rows.
6. **Delivery and newborn.** Record the pregnancy outcome. On a live birth, link the newborn as a contact-only person with a parent relationship, and create a newborn testing task.
7. **Close.** Close with a required reason. If the case is still open after delivery, staff choose whether it continues under the adult template.

Cross-links between children and adults

These are the links a separate child system and adult system cannot make. Each creates a task for a person to review, never an automatic case.

Trigger	Action
Adult occupational case; children or a pregnant person share the adult's primary residence or a relationship	Task: recommend testing them; link the records.
Child case; a household member's employment lists metal tasks	Task: occupational inquiry for that adult, even if they are a contact-only person.
Child case; other children share the primary residence or a sibling relationship	Task: recommend testing the other children.

Trigger	Action
Two or more cases at one employer worksite within the configured window	Cluster alert to the occupational team.
Two or more child cases at one location, building or child-care site within the window	Cluster alert to the residential team.
Case for a pregnant person	Task: prenatal provider notification and newborn testing follow-up.

User interface

The person page has the same sections for everyone; the population only changes their order and which empty states appear. Nothing about a person is entered into a field meant for someone else.

Person page

Section	Contents	Child order	Adult order	Pregnant order
Header	Name, DOB, age, population badge, open-case chips per analyte, current primary residence county; aliases one click away	1	1	1
Relationships	Guardians, parents, household members; "Add relationship" picks or creates a person	2	6	7
Places	Residences, child care, school, other sites; role chip, primary star, schedule, dates	3	3	4
Results	Table filtered by an analyte dropdown; trend chart per analyte with threshold lines, venous and capillary marked differently	4	4	5

Section	Contents	Child order	Adult order	Pregnant order
Cases and investigations	One card per case with its workflow steps	5	5	6
Employment	Current and past jobs, worksite, codes, metal tasks	6	2	3
Exposure sources	Non-occupational sources	7	7	8
Pregnancy	Episodes with dates, outcome and linked newborn	8	8	2
Timeline	Results, address changes, job changes and case events on one time axis	9	9	9

Empty sections show a short empty state, such as "No guardians recorded" or "No jobs recorded", instead of disappearing. A 17-year-old with a job sees the Employment section filled in with no special handling. The Pregnancy section is hidden for a person recorded as male with no episodes.

Places behavior

- "Set as primary" asks for a reason and the start date, splits the rows as address rule 12 describes, and shows the county change.
- Adding an overlapping residence opens the primary choice inline; it cannot be saved without one.
- Address entry standardizes as the user types and offers matching existing locations before creating a new one.
- A child-care or school entry is picked from organizations first; the site address fills in.

Result entry form

Fields cascade from the analyte lookup so users only see what applies:

1. Analyte drop-down from active analytes, in sort order.
2. Specimen drop-down filtered to that analyte's specimens.
3. Unit drop-down filtered to that pair's accepted units, canonical unit first.
4. Sample type shown only when uses_sample_type is true.

5. Speciation shown only when options exist.
6. Creatinine shown only when required.
7. Collection, analyzed and reported dates, value with operator, test method, lab, ordering provider, accession number.

On save the form shows the canonical value and whether any threshold was crossed.

Work queues

- Unmatched results, with suggested person matches and scores.
- Unknown LOINC codes or units.
- Batches paused because a sender's layout changed.
- Address review: new lab addresses that differ from the primary.
- Unverified primary residences on open cases.
- Results whose detection limit is above the case level, by reporter.
- Pregnancy episodes past their expected end.
- Cases with overdue steps, by worker.
- Cross-link and cluster tasks.
- Results missing extract-required fields before the next CBLS or ABLES submission.

General

- Global search by name, DOB, address, organization, accession number or case number.
- Lists export to CSV and Excel within the user's access scope, for users with the export permission; each export is audited with its row count.
- WCAG 2.1 AA, keyboard-only operation, and layouts that work at tablet width for field staff.

Intake, matching, extracts and security

Intake formats

Every format is parsed directly into one internal staging record, so no agency has to convert files to HL7 before loading them. Today some agencies do exactly that; for example, MDE converts CSV files to HL7 v2.3.1 for HHLPSS.

Format	Handling
HL7 v2.5.1 ORU^R01	Standard ELR. Ask-at-order-entry answers, such as the employer, occupation and guardian fields in LOINC panel 80399-9, are mapped when sent.
HL7 v2.3.1 ORU^R01	Still widely sent. Its own parser, with per-sender differences in segment and field use handled in the sender's profile.
eICR (CDA; FHIR when adopted)	For jurisdictions that receive lead case reports through eCR. Its own parser; carries pregnancy and occupational data that ELR often lacks.
CSV, TSV, pipe-delimited, fixed-width	Mapped by the reporter's intake profile.
Excel (.xlsx, .xls)	Any sheet and header row, mapped by the reporter's intake profile.
Web portal	Single or batch entry for clinics and point-of-care sites with no file feed.
Manual entry	Result entry form for fax and paper; the scan attaches to the result.

Files arrive by upload, secure file drop (SFTP) or a feed from the state's integration engine. Parsers are plug-ins, so a new format (FHIR, for instance) can be added without touching the rest of the pipeline. When HL7 messages arrive directly rather than through the integration engine, the system returns acknowledgments in the mode the sender's profile names.

Intake profiles

Labs ignore published templates, so intake cannot depend on them. Instead, each sender has a saved intake profile that describes its own layout.

- **What a profile holds:** file type, sheet and header row, delimiter, date formats, column-to-field mapping, value translations ("V" → venous, "M" → male), local test codes → LOINC, default units, and constants for values the file leaves out, such as the lab name.
- **Building one:** the user uploads a sample file. The screen shows each column with sample values beside the target fields, suggests matches from header names and content (dates, ZIP codes, names), and previews the first rows parsed and validated

before the profile is saved.

- **Layout drift:** each profile stores the file's header signature. When a sender's columns change, the whole batch pauses before loading and asks a user to update the mapping, instead of loading values into the wrong fields. A paused batch resumes once the mapping is fixed; it is not a failed batch.
- **Versioning:** profiles are versioned. A batch loaded with a bad mapping can be reprocessed with the corrected version.
- **Recommended template:** the project still publishes a CSV and Excel template with a data dictionary, and shows each reporter's template compliance, but nothing breaks when a reporter ignores it.

Intake pipeline

1. **Receive:** store the raw file or message unchanged.
2. **Parse:** apply the sender's profile to produce staging rows.
3. **Validate:** check each row; every problem gets a code, the field, and a plain-language message.
4. **Normalize:** convert units, codes and dates.
5. **Match:** link to an existing person, send to review, or create a new person. Guardian, household and employer fields on the report create or link contact-only persons, relationships and employments, marked unverified. A guardian value that looks like an organization goes to review as a possible employer.
6. **Commit or hold:** clean rows load; the rest are held with their reasons.
7. **Classify:** derive population, compare thresholds and run case rules for each loaded row.

Apart from a layout pause, a batch never fails whole. Each batch shows counts of received, loaded, held and rejected rows, linked to the rows themselves.

Held records

A held record is a staging row that could not load on its own. The queue is built around fixing causes, not editing rows one at a time.

- **Specific reasons.** Every hold names its reasons from a fixed, system-defined list: unknown sender, unknown test code, unknown unit, inactive analyte, missing result value, missing DOB, unusable address, possible duplicate result, uncertain person match, out of jurisdiction, or impossible dates.

- **Grouped by cause.** The queue groups holds by reason, then by sender and batch. Mapping one unknown code releases every row held for that code.
- **Fix once, remember it.** A fix can be saved back to the sender's profile ("always translate this value for this lab"), so the same problem does not recur.
- **Edit in place.** The reported value stays visible beside the correction. Corrections are logged; the result's as-reported snapshot is never changed.
- **Urgent first.** Held results at or above an action level sort to the top and appear on the home dashboard, because an elevated result sitting in a queue delays follow-up. Without a usable DOB, the lowest action level for any population applies.
- **Ownership and age.** Holds can be assigned, and the queue shows how long each has waited.
- **Release and reject.** Releasing reruns validation and matching. Rejecting requires a reason, keeps the row, and can generate a follow-up letter to the reporter.

Person matching

Each incoming record ends in one of three plain outcomes: linked automatically, needs review, or new person. Matching compares names and aliases, DOB, sex, address and external identifiers, such as a medical record number with its assigning facility.

Review screen

- The incoming record and up to five ranked candidates appear side by side, field by field.
- Each field is marked as agreeing, close (a nickname, or a DOB with day and month swapped), different or missing.
- A one-line explanation says why the candidate was suggested, such as: same last name and DOB; first name "Jon" vs. "Jonathan"; same address.
- The candidate's result history and address history are one click away.
- Three buttons: Same person, Different person, Not sure. "Not sure" sends the record to a supervisor. "Different person" is remembered, so that pair is never suggested again.

Built-in safeguards

- Placeholder names such as "BABY BOY" never link automatically.
- Same DOB and address with a different first name (possible twins) never link automatically.

- Merges made later from the person page show a preview of everything that will move, and can be undone. When both records have an open case for the same analyte, the merge combines them into the earlier case and reruns classification. A merge of records owned by different tenants goes to their nearest common ancestor.

The same review screen is used for duplicate organizations, locations and providers.

Matching configuration

Administrators choose behavior in plain words, not weights and scores.

- **Presets.** Strict, Balanced and Lenient, each described by its effect. For example, Strict links only when name, DOB and sex agree exactly, and sends more records to review. Strict is the default.
- **Plain-language rules.** An advanced view lists rules as on/off switches: treat nicknames as the same first name; allow day and month swapped in DOB; use address to break ties; ignore hyphens and suffixes in last names. Numeric scores stay hidden unless expanded.
- **Test before saving.** A proposed change runs against the last 90 days of incoming records and past review decisions. It shows how many records would now link automatically, how many would go to review, and how many past "Different person" decisions it would have linked by mistake.
- **History.** Every configuration change is versioned and audited, and can be rolled back.

Federal extracts

Both extracts are generated, validated and stored by the system on a schedule, with a validation report before submission. Names, street addresses, phone numbers, SSNs and medical record numbers never leave in either file; each carries only the fields its federal layout requires, such as date of birth, county and a stable surrogate ID.

- **CBLS (children):** quarterly ASCII files built to CDC's current variable specification. CDC requires quarterly reporting from the states it funds.
- **ABLES (adults):** an annual file built to NIOSH's current submission layout. CDC's information collection notice counts one ABLES submission per state per year, against four CBLS files.
- **Layouts are configuration.** Field order, widths, code maps and cross-field rules are versioned layout rows, so a federal layout change is a configuration update, called out in release notes.

ABLES field sources:

ABLES field	Source in this model
StateRep	Jurisdiction setting (2-letter postal code).
StateRes, CountyRes	Reporting county: primary residence on the collection date. CN or MX for Canada or Mexico; 99 or 999 when unknown.
StateExp, CountyExp	Worksite of the employment linked to the case; otherwise the exposure source's location. Left unknown unless the exposure place is known.
ID	Stable surrogate ID per person, never name or SSN; 15 characters at most.
Status	Computed with the layout's rules for the calendar year (see Adult lead).
BLLDate, DateType	Collection date, coded as date of draw.
BLL	Canonical value as a whole number, rounded by the layout's rule.
DOB, Age, Sex	Person; age computed at collection.
Ethnicity, Race1–Race8	Person ethnicity and race values, mapped to the eight race flags.
WorkRel	Case exposure category: occupational → 1, residential or other non-occupational → 2, both → 3, unknown → 9.
NAICS	Organization code, or the employment's override, in NAICS 2022; 999 when unknown.
COC	Employment occupation code, crosswalked to the layout's Census 2002 codes; 990 when unknown.
Process	Exposure source description, with the source type's ABLES code as a fallback; NA when the exposure is work-related or unknown.

The extract validator applies NIOSH's cross-field rules. For example, NAICS and COC are required when the exposure is work-related and blank when it is not.

Security and audit

- Access comes from profiles assigned at tenant nodes (see the architecture section).
- Shipped profiles: intake clerk, case manager, investigator, occupational investigator, supervisor, epidemiologist (de-identified views), administrator.
- Local health department users see persons whose current primary residence is in their jurisdiction (a county, an independent city or a health district, as configured), plus persons on cases they own or that are shared with them.
- Encryption in transit and at rest, multi-factor sign-in, and session time-outs.
- Audit log of every view, change and export of a person, result or case, searchable by user and record. Search text goes to a separate, purgeable log.
- Application and error logs carry record IDs, never names, dates of birth, addresses or result values.
- Retention and destruction follow the state records schedule; the architecture section covers how purges reach history tables and backups.

Sources

- ABLES data submission information (NIOSH)
- HHLPSS data management platform (CDC)
- About the data: blood lead surveillance (CDC)
- Childhood blood lead surveillance: state data (CDC)
- Lead in Blood 2023 case definition (CDC)
- Blood Lead Surveillance System information collection notice (Federal Register, 2024)
- NYS Heavy Metals Registry
- Maryland Childhood Lead Registry 2000 report (MDE)

Provider and laboratory reference data

Ordering providers are identified by NPI and laboratories by CLIA number, using free federal sources. Staff stop retyping names and addresses, and reports can group results by the real provider and lab rather than by spelling variants.

Sources

Source	What it provides	Access	Refresh
NPPES NPI Registry API	Individual (NPI-1) and organization (NPI-2) providers: name, taxonomy, practice address, phone	Read-only JSON over HTTPS, version 2.1 only; 1.0 and 2.0 are retired	Registry pulls from NPPES daily
NPPES data dissemination file	The full NPI file	Bulk download	Monthly
CMS Provider of Services file: Clinical Laboratories	CLIA laboratories: CMS certification number, name, address, testing services, other characteristics	CSV download and data.cms.gov API	Quarterly

NPPES warns that holding an NPI does not mean a provider is licensed or credentialed. Registry data are treated as a lookup aid, not proof of licensure.

Provider entity

A **Provider** entity holds individual practitioners, separate from Organization. Lab results link to it through ordering_provider_id.

Field	Type	Req	Rules
npi	text(10)		Passes the NPI check digit (Luhn algorithm with the 80840 prefix) before any lookup. Unique when present.
last_name, first_name, credential	text	Y (last name)	From the registry when matched; editable.
primary_taxonomy	text		Code and description from the registry.
practice_location_id	FK location		Geocoded practice address.

Field	Type	Req	Rules
organizations	child table		Practices or clinics the provider works at, with dates.
registry_status	enum		Active, deactivated, not found, not looked up.
registry_checked_at	timestamp		
local_overrides	JSON		Fields staff changed; kept through refreshes and shown beside registry values.

Organization carries `npi` (NPI-2) and `clia_number` (see Organization). A CLIA number is checked for format (two digits, the letter D, seven digits) before lookup.

Behavior

1. **At intake.** When a message or file carries an NPI or CLIA number, the system checks the local cache first, then the source. It links or creates the provider or lab, and flags reported names or addresses that disagree with the registry.
2. **Without an identifier.** Reported provider and lab names go through the same matching review as people. The review screen can search the NPI Registry by name, city and state, and staff pick the right record.
3. **Manual entry.** Typing an NPI or CLIA number fills in the name and address; typing a name offers registry matches.
4. **Refresh.** A scheduled job rechecks NPIs on recently used providers and loads each quarterly CLIA file. Deactivated NPIs and CLIA numbers missing from the latest file are flagged for review, never deleted.
5. **Point-of-care sites.** Clinics running point-of-care analyzers are linked to their CLIA record in the reporter registry. This gives a ready list of sites for device recall outreach.

Privacy and network rules

- Lookups send only an NPI, CLIA number or provider name. Patient data never leaves the system.
- Results are cached, and calls are rate-limited and retried politely.

- Some state networks block outbound calls. Each jurisdiction can turn live lookups off and load the monthly NPI file and quarterly CLIA file instead; the choice is a jurisdiction setting (see Jurisdiction configuration).

Problems other jurisdictions report

The best evidence is the 2019 CSTE assessment, in which all 50 states plus DC and New York City described their blood lead systems. Its top finding: 56% of child systems had no way to detect clusters, and nearly half found built-in reports too limited. State reports add a consistent picture of missing race, address and sample type, and point-of-care sites that report late or stop reporting.

Problem	Who reported it	Design response
No algorithm to find clusters; staff match addresses by eye	56% of child systems, 49% of adult systems (CSTE)	Cluster detection jobs (below)
Too few built-in reports; custom queries hard	46% overall; 10 of 12 HHLPSS users (CSTE)	Report library, query builder and analytics views (below)
Bugs take a long time to fix; vendor or CDC controls the code	32% overall; half of HHLPSS users (CSTE)	Open source: agencies or contractors can fix and contribute
Data updates are slow and manual	28% overall; 58% of HHLPSS users (CSTE)	Work queues, bulk actions, auto-generated tasks
No geocoding or mapping	38% (CSTE); over 10% of Colorado records lacked enough address to assign a county	Pluggable geocoder, master location table, map views
Variables missing, so data goes in notes fields	26% of child and 46% of adult systems (CSTE)	Jurisdiction-defined custom fields stored as discrete, typed data (below)
Employer typed into guardian, provider or ordering-facility fields	Several states (CSTE)	Employment and organization entities; intake flags organization-like guardian values

Problem	Who reported it	Design response
Every ELR arrives as a child and staff relabel adults by hand; adults show up in child alerts	States with integrated systems (CSTE)	Population derived from DOB; alerts and queues filtered by population
Pregnancy kept in notes, and missing from the ELR message	Several states (CSTE)	Pregnancy episodes; ELR pregnancy field mapped when sent; eICR intake
Children and adults with the same exposure linked by hand, if at all	Half of adult programs have no method; most others manual (CSTE)	Automatic cross-link tasks
No master address index; address not tied to each test; address history wrong	Several states (CSTE)	Location table independent of cases; address snapshot on every result
Home risk assessments stored as Word documents	One state (CSTE)	Structured risk assessment forms and sample tables; files attach but do not replace fields
Labs report by fax, Excel, flat files and portals; only some by HL7	HL7 averaged 61% of reports (CSTE); Kentucky keys out-of-state labs by hand	Per-reporter intake profiles and a web portal for small reporters
Point-of-care sites report sporadically or stop	North Carolina	Reporter registry with expected cadence and silence alerts (below)
Recall retesting hampered by incomplete device-user lists and bad data	States responding to the LeadCare recall	Test method, device and kit lot on each result (below)
Race missing for most records	72% of tested children in Tennessee (2015–2022); about two-thirds in Pennsylvania	Completeness dashboard; enrichment from vital records with provenance (below)
Sample type missing	About 19% of Washington tests (2018–2022)	Completeness tracking by reporter; follow-up letters

Problem	Who reported it	Design response
Local health departments can't use the state system, or their data drifts out of sync	Several states (CSTE)	One install with local users and jurisdiction-scoped access
Unclear whether state or local staff own adult vs. child cases	One state (CSTE)	Case ownership rules (below)
Lead lives inside NBS or Maven and can't easily be replaced	A third of child systems (CSTE)	Standard interfaces to sit beside a communicable disease system (below)
Cost and state IT approval block adoption	92% need funding and 75% must meet state IT rules before switching (CSTE)	Free license, containers, security documentation

Dependence on one federal team is a real risk, not a theoretical one. CDC's Lead Poisoning Prevention and Surveillance Branch was cut in the April 2025 HHS reduction in force, and HHS rehired the team in June 2025 (Fierce Healthcare).

Cluster detection

Scheduled jobs flag groups of cases for staff review. Groups are formed by:

- shared location, building or child-care site
- shared employer or worksite
- shared ordering provider
- shared consumer product
- cases within a set distance of each other

Each rule has a configurable time window and minimum count. A flagged group becomes a cluster record that cases join or leave, with its own notes and status. Nothing is merged automatically.

Reports and analytics

- A library of standard reports: testing counts and rates, elevated counts by population and analyte, case timeliness, investigation outcomes, reporter timeliness and completeness.

- A query builder for ad hoc line lists, with saved queries shared across users.
- Read-only analytics views, with de-identified versions, published to a per-root-tenant analytics database refreshed on a schedule. Epidemiologists use SAS, R, Python or Tableau against that copy, never against the shared transactional database (see Row-level security).
- Every list and report exports to CSV and Excel, for users with the export permission.

Custom fields

Jurisdictions can add typed fields (text, number, date, picklist) to person, result, case, investigation and interview forms without code changes. Custom fields are stored as discrete data, appear in the query builder and exports, and are versioned like other configuration.

Reporter registry and compliance

Each reporting lab, clinic and point-of-care site is an organization with its intake profile, contact, devices and expected reporting cadence. The system alerts staff when a reporter goes silent past its cadence, and shows per-reporter timeliness and field completeness so staff know whom to call. Timeliness uses the collected, reported and received dates on each result.

Test method, device and lot

Each result records method (for example, ICP-MS, graphite furnace AA or point-of-care analyzer), device model and serial when known, and test kit lot, in the lab result fields. When a device or lot is recalled, a query returns every affected result and patient for retest outreach. Method also decides whether a venous result can confirm a case.

Data completeness and enrichment

- A dashboard tracks missing race, ethnicity, address, sample type, test method and DOB by reporter and by month.
- Matches against vital records, Medicaid or an immunization registry can fill missing fields. Each filled value records its source and date, and never overwrites a value the reporter sent.
- Reason for test and insurance or Medicaid status are captured when sent, since several states named them as missing.

Product registry

Consumer products found during investigations (spices, remedies, cosmetics, cookware, ceremonial powders) are records with brand, product name, country of origin and lab results. Several cases can link to one product, which supports cluster detection and referrals to regulators.

Case ownership rules

A rule table sets which agency and team owns a case by population, analyte, exposure category and jurisdiction. For example, local health departments own child cases and the state owns occupational adult cases. A local user who starts a case outside their ownership triggers a notice to the owner.

Working beside an existing disease system

Many states will keep NBS or Maven for other diseases. The system can receive ELR and eICR forwarded by a state integration engine, and exchange cases and results through its REST API. This lets a state adopt it for metals without replacing its communicable disease platform.

Sources for this section

- 2019 Blood Lead Surveillance System Assessment (CSTE)
- Tennessee vital records matching for race data (CDC)
- Pennsylvania 2014 lead surveillance annual report
- Washington child blood lead dashboard data notes
- Evaluating Colorado's lead poisoning surveillance system (CSTE 2015)
- Kentucky data resource guide 2024
- North Carolina point-of-care reporting requirements
- LeadCare recall response interviews (CDC)

Architecture: tenancy, authorization and data access

One installation serves many agencies. Each agency is a node in a tenant tree, users get permissions through profiles assigned at a node, and the database itself filters rows, so a bug in application code cannot expose another tenant's records.

Stack

- ASP.NET Core on the current .NET long-term support release, serving a REST API documented with OpenAPI. The same API serves the user interface and outside integrations.
- An Angular front end in TypeScript, served from the same origin as the API, calls the REST API through a client generated from the OpenAPI document. Sign-in uses secure, HTTP-only session cookies with CSRF protection, so no access tokens are kept in browser storage.
- Dapper for data access. Every read and write goes through a stored procedure; the application's database login has EXECUTE on procedures only, with no direct table access.
- SQL Server 2022 or later is the baseline database. PostgreSQL may come in a later release under the conditions in Database support below.

Front end

Angular and TypeScript render every screen. USWDS supplies the look (CSS, markup patterns and design tokens) where it fits, and Angular with the Angular CDK supplies all interactive behavior.

- **Generated API client.** The TypeScript client is generated from the API's OpenAPI document during the build and never edited by hand. A build fails if the client and the API disagree.
- **Same origin.** ASP.NET Core serves the compiled Angular files from the same host as the API. No CORS configuration is needed, session cookies are SameSite=Strict, and a Content Security Policy limits scripts to the application's own files.
- **USWDS without its JavaScript.** USWDS styles and tokens are used; its JavaScript is not. Components that depend on it, such as the combo box and date picker, are rebuilt with Angular and the CDK using USWDS markup and classes.
- **Data-heavy screens.** USWDS tables do not provide server-side paging, filtering or bulk actions. Held records, work queues, results, line lists and query output use one shared data grid built on the Angular CDK table and virtual scrolling, styled with USWDS tokens so it matches the rest of the application.
- **Accessibility.** USWDS patterns plus the CDK's focus and keyboard tools, checked with automated accessibility tests on each build and manual testing before release.

Screen type	Approach
Forms: person, result entry, case, investigation, interview	USWDS form markup and styles; Angular reactive forms for validation
Header, navigation, alerts, banners, step indicators	USWDS components; behavior in Angular
Held records, work queues, results, line lists, query output	Shared data grid: server-side paging, sorting and filtering, column chooser, saved views, bulk selection, keyboard navigation
Dialogs, menus, combo boxes, date pickers	Angular CDK overlays and accessibility tools, with USWDS styling
Matching review, person timeline, maps	Custom Angular components using USWDS tokens

Shared database

All tenants, including separate root tenants (different states), live in one shared database. Isolation rests entirely on tenant columns and row-level security, so the design has to make that airtight and fast.

- **Tenant columns on every row.** Every tenant-owned table carries `root_tenant_id` and `owning_tenant_id`. The root column is denormalized onto every row so the security predicate never needs a join to find it.
- **Person access table.** Persons and results are owned by the root but seen by jurisdiction. A person-access table lists the tenants that can see each person, maintained by the procedures that change a primary residence or a case's ownership or sharing, so the predicate is one indexed lookup and never joins to addresses.
- **Indexes lead with the root.** Clustered and main nonclustered indexes start with `root_tenant_id`, which keeps each state's rows together and keeps predicates cheap.
- **Keys.** Primary keys are UUIDs generated by the application, with a time-ordered part placed where SQL Server sorts first and a random part. Inserts stay in index order, yet IDs can't be guessed or used to count records. NEWSEQUENTIALID is not used, because Microsoft warns that its next value can be guessed.
- **No tenant-specific schema.** Jurisdiction differences live in configuration and custom-field rows, never in per-tenant tables or columns.

- **Per-tenant export and removal.** Backups cover the whole database, so the project ships procedures to export one root tenant's complete data set and to purge it, including its rows in temporal history tables. This covers a state that leaves, or a records-retention request. The purge report states when the data will age out of backups.
- **Fair use.** Batch jobs (intake, cluster detection, extracts) run in per-tenant queues with limits, so one state's large file cannot starve the others.
- **Audit and history.** SQL Server temporal tables keep row history, and ledger tables make the audit log tamper-evident. Transparent data encryption protects data at rest. Ledger history cannot be deleted, so the audit ledger holds only IDs, actions, users and times; names, dates of birth, addresses, result values and search text stay out of it.

Database support

SQL Server 2022 is the minimum version and the reference implementation: every feature ships and is tested on it first. Later versions, such as SQL Server 2025, are supported once the cross-tenant test suite passes on them. PostgreSQL may be added in a later release, once SQL Server support is stable, adopters request it, and it does not affect SQL Server support. It would be a separate implementation behind the same interface.

Concern	SQL Server 2022 (baseline)	PostgreSQL (optional)
Stored logic	T-SQL stored procedures	PL/pgSQL functions returning tables, with the same names and parameters
Request context	`sp_set_session_context` with read-only values	`set_config(..., true)` scoped to the transaction
Row-level security	Security policies with filter and block predicates, including on history tables	Row security policies; tables forced to apply them, and the application role never owns tables
History and audit	Temporal tables, ledger tables	History tables kept by triggers; audit hashing in the application
Overlap rule for primary residence	Checked in the procedure with row locks	The same check; an exclusion constraint may be added

Rules that keep PostgreSQL from affecting SQL Server:

1. The C# code talks to a data-access interface. A SQL Server implementation and an optional PostgreSQL implementation each provide their own procedures and Dapper calls.
2. Schema and procedures live in separate folders per database, with their own migration scripts.
3. The shared cross-tenant security test suite runs against both. PostgreSQL failures never block a SQL Server release.
4. No feature is held back or simplified to suit PostgreSQL. If something cannot be matched there, PostgreSQL documents the gap.

Tenant tree

- A tenant is an organization that uses the system: a state agency, a program inside it, or a local health department.
- Tenants form a tree of any depth. Root tenants are fully isolated from each other; no data crosses between roots.
- Ancestry is kept in a closure table (ancestor, descendant, depth), so "this node and everything below it" is one indexed join.
- Each tenant maps to one or more sub-jurisdictions (counties, independent cities), which routes new cases.

Example of one state's tree:

Tenant	Parent	Typical work
State lead program	none (root)	Intake, matching, federal extracts, statewide view
Occupational health program	State lead program	Adult occupational cases
County A health department	State lead program	Child cases for County A residents
City B health department	State lead program	Child cases for City B residents

A city that wants full independence runs as its own root, or as its own install; the state then receives that city's data through extracts or the API, not shared records.

Data ownership

Data	Owner	Visible to
Shipped reference data: analytes, LOINC maps, NAICS and Census codes, extract layouts	System	Every tenant, read-only
NPI and CLIA cache	System	Every tenant
Configuration: thresholds, workflows, matching	Root tenant; child tenants override only what the root allows, such as letter templates	The tenant and its descendants
Persons and results	Root tenant	Users whose scope covers the person's current primary residence, or a case they can see
Cases, investigations, activities	Tenant set by the case ownership rules	Owning tenant, its ancestors, and tenants it is shared with
Locations, organizations, providers	Root tenant	Every user under that root

Persons stay one record statewide, so a family moving between counties does not create duplicates; cases are owned locally.

Profiles and permissions

- **Permission:** an action on a resource, such as person.view, person.view_identifiers, result.release_hold, case.edit, data.export, matching.configure, extract.submit or tenant.administer.
- **Profile:** a named bundle of permissions, such as "Local case manager". Profiles are defined at the root and inherited down the tree. A tenant administrator cannot grant permissions they do not hold.

- **Assignment:** user + profile + tenant node + scope (this node only, or this node and everything below), with start and end dates.
- A user can hold several assignments, for example statewide epidemiologist and county case manager.
- Effective permissions are resolved at sign-in and cached for the session; an assignment change invalidates the cache.
- Administrator profiles do not include case editing by default, to separate duties.

Row-level security

1. On every request, the application opens a connection and calls `sp_set_session_context` with the user ID and a session scope key, marked read-only so nothing later in the request can change it. The key points to the user's resolved tenant set in a protected scope table, so predicates test membership with an index seek instead of parsing a list.
2. SQL Server security policies apply filter predicates (rows outside the user's scope are invisible) and block predicates (rows cannot be inserted or moved into a tenant outside scope) on every tenant-owned table. Each temporal history table gets its own policy, because row-level security predicates are not copied from a current table to its history table.
3. Person and result predicates read the person-access table, so a jurisdiction check never joins to addresses.
4. Stored procedures also check permissions. Row-level security is the second wall, not the only one.
5. Pooled connections get fresh context on every open. Predicates fail closed when context is missing.
6. De-identified profiles read through views that drop identifiers.
7. No profile runs ad hoc SQL against the shared database, because crafted queries can infer filtered rows through errors. Analysts use the per-tenant analytics database instead.
8. An automated test suite attempts cross-tenant reads and writes through every procedure on each build, including history tables and the analytics publish job.

Cross-tenant work

- **Case transfer:** when a child moves to another county, the case moves by a transfer request the receiving tenant accepts. History stays with the case.

- **Cross-tenant links and clusters:** tasks that span tenants go to their nearest common ancestor, often the state program, which can share the case with each tenant involved.
- **Search outside scope:** returns only "a record exists in another jurisdiction", without details, unless the user holds a profile that covers it. The check runs only on an exact name and DOB match, through one procedure that reads a match-key table outside row-level security, and every hit is audited.

Authentication

Every user signs in with multi-factor authentication. Built-in accounts are the default; Microsoft Entra ID is an option an adopter turns on.

Built-in accounts (default)

- User ID and password, stored with ASP.NET Core Identity's password hasher. Passwords are never logged or emailed.
- Password rules follow NIST SP 800-63B-4 (July 2025) by default: a minimum of at least 8 characters, which NIST permits only because every account has a second factor; a check against known breached passwords; no composition rules; and no forced rotation. Agencies whose policy requires rotation or complexity rules can turn them on, and the settings screen notes that NIST now says not to.
- Multi-factor authentication is required for every user and cannot be turned off. Users enroll at first sign-in with an authenticator app (time-based codes) or a security key or passkey (FIDO2/WebAuthn). Administrator profiles require a security key or passkey, because those resist phishing.
- Email codes and text-message codes are two separate settings, both off by default. Under 800-63B-4, email may not be used for out-of-band authentication and text messages are a restricted authenticator, so the settings screen says so before either is enabled. An adopter can enable either one, both, or neither, since some states allow email codes but not text messages.
- One-time recovery codes are issued at enrollment. Accounts lock after repeated failed attempts, and a tenant administrator can reset a user's MFA after verifying the person.

Microsoft Entra ID (optional)

- Enabled by configuration per root tenant, using OpenID Connect. Nothing about Entra is required for adopters who do not turn it on. The sign-in code is standard OpenID Connect, so another provider can be added by configuration later; Entra ID is the one the project tests.

- Users sign in through Entra, so the agency's own MFA and conditional access policies apply. If the sign-in token does not show that MFA was performed, the application asks for its own second factor.
- Entra users are linked to application users by their Entra object ID. Administrators add them by hand or through SCIM provisioning from Entra.
- Both methods can run side by side, for example state staff on Entra and local health department staff on built-in accounts.

For both methods

- Sign-in only establishes who the user is. Tenant assignments and profiles still decide what they can see and do.
- Sign-in method, MFA options and password rules are tenant settings, recorded and audited like other configuration.
- Sessions time out after inactivity, and tokens never carry patient data.

Jurisdiction configuration

An adopting agency sets these at install and can change them later through the admin screens. Each change is effective-dated and audited.

Setting	What the jurisdiction decides	Shipped default
Jurisdiction identity	State postal code, state FIPS, agency name, letterhead	None; required at install
Sub-jurisdictions	Counties, independent cities or health districts, and which counties map to each	The state's county list
Access model	The tenant tree: programs and local health departments as child tenants, or a standalone local root	State with local users
Reportable analytes	Which metals and specimens are reportable, and at what levels	Blood lead active at all values; others present but inactive

Setting	What the jurisdiction decides	Shipped default
Case and action levels	Child, adult and pregnancy case levels, action tiers and due intervals	CDC child blood lead reference value; federal adult rows shipped inactive; adult levels must be set
Confirming methods	Which test methods confirm a venous result, and whether an unreported method does	Graphite furnace AA and ICP-MS; unreported method confirms
Adult age cutoff	16, 18 or another age for workflows	16, matching ABLES and the national case definition
Pregnancy episode length	How long an episode with no end date stays open	42 weeks from start
Primary residence policy	Whether a new lab address moves the primary or goes to review	Review
Investigation lookback	How far back past addresses and jobs count	Must be set at install
Site investigation minimum	Weekly hours at a relative or sitter home or other regular site that trigger an investigation request	Must be set at install
Workflow templates	Steps, owner roles, due-date rules and closure criteria per population	Child residential, adult occupational and pregnancy templates
Contact attempts before a letter	Failed contact attempts before a letter is queued	Must be set at install
Local notification	Whether opening a case notifies the local health department	Off
Medical removal and return-to-work levels	Levels for adult retest tracking	None; loaded from the OSHA or state-plan standard

Setting	What the jurisdiction decides	Shipped default
Arsenic handling	Whether a high total urine arsenic opens a case or a "speciation recommended" task	Task
Occupational referral agency	Federal OSHA area office or the state-plan agency	Federal OSHA
Cluster rules	Time window, minimum count and distance for each cluster rule	Must be set at install
Letters	Templates, languages, signatures	English samples
Federal extracts	CBLS and ABLES on or off, schedule and layout version	CBLS quarterly and ABLES annually, both on
Matching preset	Strict, Balanced or Lenient, plus rule switches	Strict, sending more to review
Provider and lab lookups	Live NPPES and CMS lookups, or monthly and quarterly file loads	Live lookups
Sign-in	Built-in accounts, Entra ID or both; email and text-message codes; password rules	Built-in accounts; email and text codes off; NIST 800-63B-4 password rules
Records retention	Retention periods by record type, from the state records schedule	Must be set at install

Adoption and open-source considerations

Adoption depends less on features than on how cheaply an agency can move its existing data in and pass its own IT review.

- **Migration toolkit.** HHLPSS has no export feature, so each adopter builds its own migration. The project provides documented import staging tables, CSV templates and an import API, keeps a legacy ID on every imported record for traceability, and flags

guardian values that look like employers for review.

- **Deployment.** One installation can host a state and its local health departments as a tenant tree, or an agency can run its own install, on its own cloud tenancy or servers. The project publishes container images. No dependency on a hosted service run by the project.
- **Reference data shipped with the code.** Analyte and LOINC mappings, NAICS and Census occupation code lists with their versions and crosswalks, county FIPS codes, federal extract layouts, and federal levels tagged with their authority and shipped inactive. State-specific levels are never shipped as defaults.
- **Standards in and out.** HL7 v2.5.1 ELR and eICR for intake, CDC and NIOSH layouts for extracts, and a documented REST API. A FHIR interface can follow.
- **No real data in the repository.** A synthetic data generator produces demo and test records, including shared-custody children, adults with several jobs and take-home exposure households.
- **Security documentation.** A controls mapping (for example, to NIST SP 800-53) and a data-flow diagram, which state IT security reviews usually ask for.
- **Accessibility.** WCAG 2.1 AA and Section 508 conformance, with an accessibility statement. WCAG 2.1 AA is also the standard in DOJ's ADA Title II web rule for state and local governments, now due April 26, 2027 for those serving 50,000 or more people and April 26, 2028 for smaller ones. HHS's Section 504 rule applies it to recipients of HHS funds, now due May 11, 2027 for those with 15 or more employees (EDUCAUSE).
- **Project basics.** The Apache License 2.0, a contributing guide, issue templates, a public roadmap, and release notes that call out any change to extract formats, layouts or classification rules.

Decisions

Topic	Decision
License	Apache License 2.0. Anyone may use, modify and distribute it, including commercially. It comes with no warranty, and redistributions must keep the copyright notice and the project's NOTICE file, which credits the source.

Topic	Decision
HHLPSS migration	HHLPSS has no export feature. Adopters build their own migration onto the project's import staging tables, CSV templates and import API.
Other exposure markers	Deferred to a later release if adopters request them. The analyte model already allows adding them.
Case definition	Child and adult classification follows CDC's 2023 Lead in Blood case definition; levels and confirming methods are settings.
Technology	ASP.NET Core, Dapper and stored procedures, with multi-tenant authentication and authorization: a hierarchical tenant tree, profile-based role assignment and row-level security.
Front end	Angular with TypeScript, served from the same origin as the API, using a client generated from OpenAPI. USWDS for CSS, markup and design tokens where it fits; interactive behavior in Angular and the Angular CDK; a shared CDK data grid for paged, filtered screens.
Database	SQL Server 2022 or later is the baseline. PostgreSQL may be added in a later release once SQL Server support is stable, adopters request it, and it does not affect SQL Server support.
Tenant isolation	All tenants share one database, isolated by tenant columns and row-level security. Analytics run on a separate per-tenant copy.
Sign-in	Built-in user ID and password with required multi-factor authentication. Microsoft Entra ID sign-in is optional and enabled by configuration, through standard OpenID Connect.

Open questions

- [ ] **Offline field entry.** Do investigators need to record samples and interviews without a connection? If so, which screens, and how are conflicts resolved on sync?
- [ ] **Service levels.** What availability, recovery point and recovery time objectives apply to a shared multi-state install, and who operates it?
- [ ] **Unreported test method.** Is counting a venous result with no reported method as confirmed acceptable for CBLS, or should CBLS confirmation require a method?
- [ ] **Adult default.** Should the shipped adult case row be the 2023 national definition (3.5 µg/dL venous) or NIOSH's 5 µg/dL level, or should both stay inactive?
- [ ] **Install-time defaults.** Should the project ship starting values for the site investigation minimum, contact attempts before a letter, investigation lookback and cluster rules?

Revision notes

The October 24, 2026 revision fixes internal contradictions, closes rule gaps that would misclassify results, tightens tenant isolation and brings federal references up to date. Each row names the section changed and the reason.

Section	Change	Why
Purpose, glossary, configuration	ABLES extract is annual; CBLS stays quarterly	CDC's 2024 information collection notice counts one ABLES submission per state per year; the spec said quarterly for both.
Design principles	Added principle 7 (automation proposes, staff decide); principle 6 covers results, cases and exports	Matches how cross-links, clusters and address changes already worked, and the security section's audit scope.
Data model overview	18 core entities; Exposure source and Provider added to the diagram; supporting tables listed	Both entities were defined but missing from the 16-entity overview.
Entity definitions	Enumerations split into system-defined and administrator-maintained	A fixed list of hold reasons contradicted "every enumeration is an admin lookup".

Section	Change	Why
Person	Sex adds other; race maps to all eight ABLES flags; external identifiers; death creates a task instead of closing cases	ABLES Race1–Race8 are fixed category flags (including refused and unknown), not eight free slots, and ABLES sex has an other code.
Person alias	Field table added	The entity had no fields.
Pregnancy episode	End rule, outcome field; male sex warns instead of blocking	An episode with no end date stayed open forever; blocking rejects valid reports.
Person relationship	Direction defined; sibling role; verified_date	"Parent" did not say which side was the parent.
Organization, Provider	npi added to Organization; duplicate clia_number removed from the provider section; NAICS version stored	The provider section re-added a field Organization already had.
Location	Country and non-US fields, city-only locations, manual pins, building record defined, owner history, census vintage	US-only addresses blocked cross-border and homeless records; building_id pointed at nothing; owners change during orders.
Employment	industry_text, verified_date, occupation code version	ABLES expects Census 2002 occupation codes; coding needs the stated industry.
Exposure source	Casting and pica types; ABLES Process mapping; product link	Aligns with ABLES Process codes and the product registry.
Lab result	Result status and corrections; reported date vs received time; method, device, lot, reason and payer fields; ordering provider links to Provider	Lab corrections were treated as duplicates; method decides confirmation; fields described elsewhere were missing from the table.
Investigation	Targets a place, a worksite or a source; links to many cases; reuses open investigations	One case per investigation duplicated work for siblings and co-workers; product investigations had no target.

Section	Change	Why
Environmental samples	Effective-dated standards table; computed exceedance; instrument file import	EPA's 2024 dust rule changed the standards; a typed flag cannot follow such changes.
Address rules	Reporting county vs assignment county; hours threshold for other places; row splitting; no fixed address	The example skipped the grandmother's home though the rules said every place is investigated; follow-up belongs where the family lives now.
Thresholds	Action rows keyed by action code, with due intervals, optional age band and sample type; reference rows inactive	The overlap rule blocked tiered action levels such as CDC's confirmation schedule.
Normalization	Population and thresholds run after matching; below-detection rule	Population needs the matched person's DOB and pregnancies; a "<5" result cannot be judged against 3.5.
Child lead	Cites the 2023 Lead in Blood definition; confirming methods; unknown sample type; annual new-case rule	The spec cited the 2010 definition; the 2023 definition requires GFAAS or ICP-MS for venous confirmation.
Adult lead	Unknown sample type counts as venous; ABLES Status uses NIOSH's 10 µg/dL rule	ABLES status codes use their own line, not the adult case level.
General rules	Open cases keep classification on a threshold change; outreach report	The spec covered closed cases only.
Investigation workflows	Pregnancy template; closure criteria as settings; guardian from the lab report; shared workplace investigations	Pregnant cases had no template.
Cross-links	Testing tasks for siblings and pregnant household members	Common follow-up the table missed.

Section	Change	Why
User interface	Pregnancy section and order column; export permission; new queues	Pregnancy episodes had no place on the person page.
Intake	eICR format; HL7 acknowledgments; classify step after commit; guardian and employer links from reports; layout pause defined	Layout drift stopped whole batches while the text said a batch never fails whole.
Matching	External identifiers; merge rules for open cases and cross-tenant records; preset names match configuration	Merging two open cases broke the one-open-case rule; the configuration table described scores.
Federal extracts	Direct identifiers named; BLL field added; field mappings and codes; layouts as configuration	Both files carry DOB, so "personal identifiers never leave" was inaccurate.
Security and audit	Local access by current residence or case; search text and application logs kept free of patient data	Aligns with Data ownership; ledger history cannot be purged.
Architecture	Person-access table; session scope table; history-table policies; analytics copy; UUID guidance; ledger without patient data	History tables do not inherit row-level security, ad hoc SQL can leak rows through errors, NEWSEQUENTIALID is guessable, and ledger history cannot be deleted.
Authentication	NIST SP 800-63B-4 rules; email and text-message status; phishing-resistant MFA for administrators; standard OpenID Connect	800-63B-4 (July 2025) bars email for out-of-band authentication and restricts text messages.
Jurisdiction configuration	11 settings added; matching row renamed	Settings referenced elsewhere had no row.
Adoption	Container images; DOJ and HHS accessibility dates	DOJ moved the Title II date for large entities to April 26, 2027.

Section	Change	Why
Decisions, open questions	Case definition, database and sign-in rows updated; five open questions	"None at this time" hid decisions still to make.
Sources	2010 case definition replaced with the 2023 Lead in Blood definition; extract and standards sources added	The 2010 definition is superseded.